

Inferential Statistics

Statistics and Data Analysis

Unit-3

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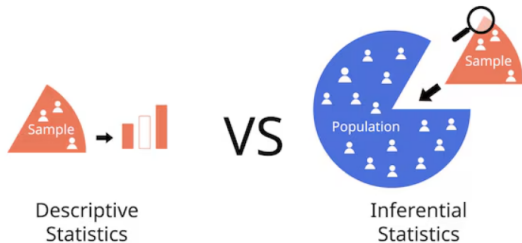
- 1 Inferential Statistics
- 2 Population and Sample
- 3 Quiz: Inferential Statistics
- 4 Sampling Techniques in Inferential Statistics
- 5 Probability Sampling
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- 7 Central Limit Theorem
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- 13 Analysis of Variance (ANOVA)

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Where Are We Now?

So far, we have focused on:

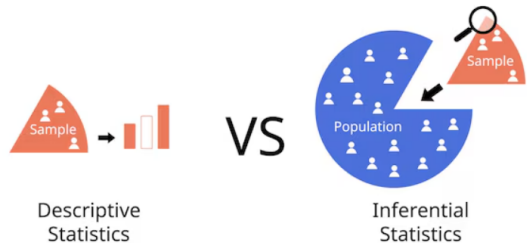
- Organizing data
- Summarizing data
- Visualizing data



Where Are We Now?

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- Organizing data
- Summarizing data
- Visualizing data



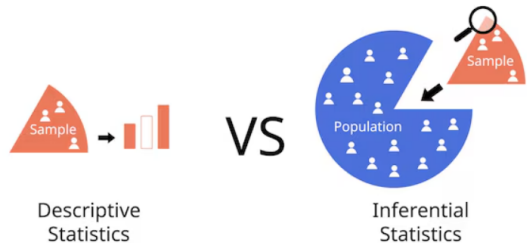
What Comes Next?

Making decisions about an entire population using limited data

Where Are We Now?

So far, we have focused on:

- Organizing data
- Summarizing data
- Visualizing data



What Comes Next?

Making decisions about an entire population using limited data

This is Inferential Statistics

Descriptive Statistics

- Describes given data
- Mean, median, variance
- Tables and graphs
- No generalization
- What does the data say?

Descriptive Statistics

- Describes given data
- Mean, median, variance
- Tables and graphs
- No generalization
- What does the data say?

Inferential Statistics

- Uses sample data
- Estimates population values
- Tests claims and assumptions
- Involves uncertainty
- What does the data imply?

Descriptive describes the sample; Inferential concludes about the population

Descriptive Statistics		Inferential Statistics	
Measures of Central Tendency	Measures of Dispersion	Hypothesis Testing	Regression Analysis
Mean	Range	Z test	Linear Regression
Median	Standard Deviation	F test	
Mode	Variance	T test	
	Absolute Deviation		

Real-Life Motivation

A university has **10,000 students**. You want to find the **average daily screen time**.

- Is it possible to ask every student?
- Is it practical to collect full data?

No

No

Solution

- Select a smaller group of students
- Analyze their data
- Draw conclusions about all students

Definition

Inferential Statistics is the branch of statistics that uses **sample data** to make **estimates, predictions, or decisions** about a **population**.

Key Characteristics

- Uses probability concepts
- Deals with uncertainty
- Generalizes from sample to population
- Forms the basis of hypothesis testing

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Definition

Population is the complete set of all individuals, items, or observations of interest in a statistical study.

Examples

- All Indian voters in an election study
- All products manufactured in a factory
- All students enrolled in a university
- All possible inputs to a machine learning model

- Population size is usually very large
- Complete data collection is costly
- Often time-consuming
- Sometimes impossible to observe fully

Key Observation

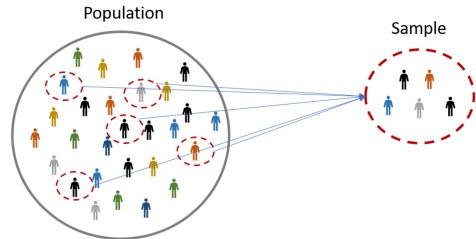
Studying the entire population is ideal, but rarely practical in real-world problems.

Definition

Sample is a subset of the population selected for analysis to draw conclusions about the population.

Example

- Population: 10,000 university students
- Sample: 200 selected students
- Conclusions are drawn for all 10,000 students



- Reduces time and cost
- Easier data collection
- Faster analysis and decisions
- Practically feasible
- Allows use of statistical models

Population	Sample
• A population is the entire collection of subjects affected by your research question.	• A sample is a subset of the population you study.
• Measurements taken from a whole population are called parameters.	• Measurements taken from a sample are called statistics.
• Data for an entire population is often very difficult or impossible to collect.	• When population data is unavailable, we use sample data to make inferences about the population.
• If you do have data for a whole population, your parameters will be "true" measures of some population characteristic.	• Sample data yield statistics, which can be used to estimate population parameters. These estimates will always involve some margin of error due to sampling bias and other errors.

Key Idea

A well-chosen sample can represent the population accurately.

Aspect	Population	Sample
Size	Large	Small
Coverage	Entire group	Part of group
Cost	High	Low
Time	More	Less
Practicality	Difficult	Feasible

Population Parameters

- Describe population
- Usually unknown
- Fixed values
- Examples: μ , σ , p

Sample Statistics

- Computed from sample
- Known values
- Random in nature
- Examples: $\bar{x}$, s , $\hat{p}$

Name	Population Parameters	Sample Statistics
Mean	μ	$\bar{X}$
Median	η	$\tilde{X}$
Mode	No symbol	No symbol
Range	R	R
Variance	σ^2	s^2
Standard Deviation	σ	s
Sample Size	N	n
Estimates	$\hat{\sigma}$	n/a

Critical Warning

A poorly chosen sample leads to wrong conclusions, even if calculations are correct.

Implication

- Sample selection is crucial
- Leads to sampling techniques
- Forms foundation of inference



- Inferential statistics draws conclusions beyond data
- Population represents the entire group
- Sample is a subset of the population
- Sample statistics estimate population parameters
- Quality of inference depends on sample quality

Next Topic: Sampling Techniques

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- ❶ Which of the following statements best explains why inferential statistics relies on probability theory?
- ❶ A Population data is always incomplete
 - ❶ B Sample statistics are deterministic values
 - ❶ C Sample outcomes vary due to randomness
 - ❶ D Population parameters change over time
- ❷ A sample mean is considered a random variable because:
- ❷ A It is calculated using random formulas
 - ❷ B Different samples yield different mean values
 - ❷ C Population parameters are unknown
 - ❷ D Sample size is usually small

- 1 Two independent random samples drawn from the same population produce significantly different statistics. This situation:
 - A Invalidates inferential statistics
 - B Indicates sampling bias
 - C Is expected due to sampling variability
 - D Implies incorrect population assumptions

- 2 Which condition is **most critical** for a sample to provide reliable inference about a population?
 - A Large sample size
 - B Random selection
 - C Low computational complexity
 - D High measurement precision

- ❶ If a sample size is increased while the sampling method remains biased, the resulting inference will:
- ❶ A Become unbiased automatically
 - ❶ B Have reduced variability but remain biased
 - ❶ C Eliminate both bias and variance
 - ❶ D Be statistically invalid
- ❷ A population parameter is treated as a fixed quantity because:
- ❷ A It is always known
 - ❷ B It depends on the sample size
 - ❷ C It does not vary across samples
 - ❷ D It changes with each observation

- 1 Which of the following scenarios eliminates the need for inferential statistics?
- A Very large random sample
 - B Small but representative sample
 - C Complete population data available
 - D Highly accurate statistical model
- 2 In inferential statistics, the primary role of a sample statistic is to:
- A Replace population parameters
 - B Describe only the observed data
 - C Estimate unknown population parameters
 - D Eliminate uncertainty in analysis

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Real Problem

In practical situations, studying the **entire population** is:

- Expensive
- Time-consuming
- Sometimes impossible

Examples

- All voters in India
- All manufactured bulbs in a factory
- All students in a university



Solution

Select a **sample** and make inference about the population.

Population

Entire group of individuals or observations under study.

Population

Entire group of individuals or observations under study.

Sample

Subset of the population selected for analysis.

Population

Entire group of individuals or observations under study.

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Subset of the population selected for analysis.

Parameter vs Statistic

Population Mean = μ

Sample Mean = $\bar{x}$

Population

Entire group of individuals or observations under study.

Sample

Subset of the population selected for analysis.

Parameter vs Statistic

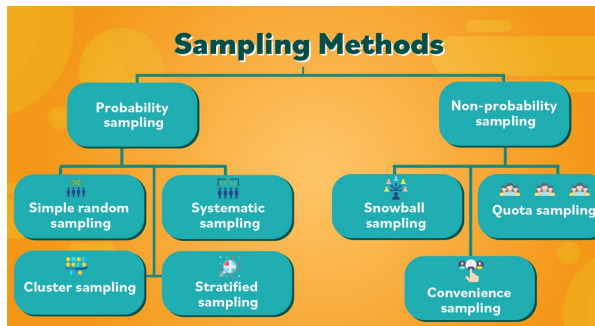
Population Mean = μ

Sample Mean = $\bar{x}$

Key Idea

Parameter is fixed but unknown.
Statistic is computed from sample.

Probability Sampling	Non-Probability Sampling
Known selection probability Used in inference Less bias	Unknown selection probability Used in exploratory research More bias



Important

Inferential statistics mainly relies on **Probability Sampling**.

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Definition

Each unit in the population has an **equal chance** of being selected.

Methods

- Lottery method
- Random number tables
- Computer-generated random numbers

Advantages

- Simple and unbiased
- Foundation of statistical theory

Disadvantages:

Not practical for large populations
Requires full list of population



Procedure

- 1 Select a random starting point
- 2 Choose every k^{th} element

Example

Population = 1000, Sample = 100

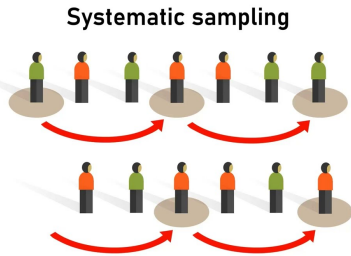
$$k == \frac{N}{n} = \frac{1000}{100} = 10$$

Select every 10th item.

Advantage: Simple and fast

Limitation

Fails if there is periodic pattern in data.



Idea

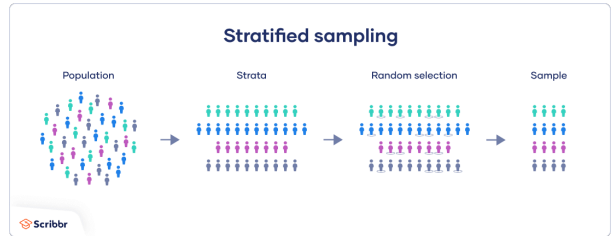
Divide population into **homogeneous groups (strata)**.

Example Students divided into:
UG, PG, and PhD

Random sample taken from each group.

Types

- Proportionate Stratified Sampling
- Disproportionate Stratified Sampling



Advantage

Reduces sampling variance.

Idea

Divide population into **natural clusters**.

Example

Select 5 schools randomly and survey all students in selected schools.

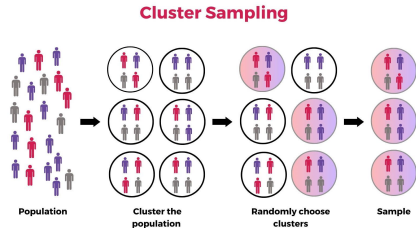
When Used

- Large geographical areas
- Cost constraints

Note

Less precise than stratified sampling.

TCMResearch



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Definition

Select individuals who are easiest to access.

Example

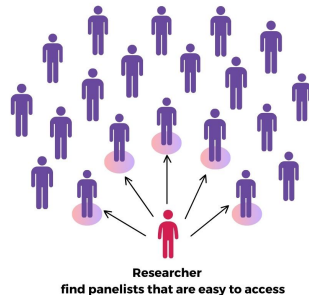
Survey students present in classroom.

Limitation

Highly biased and not suitable for hypothesis testing.

TCM Research

Convenience Sampling



Definition

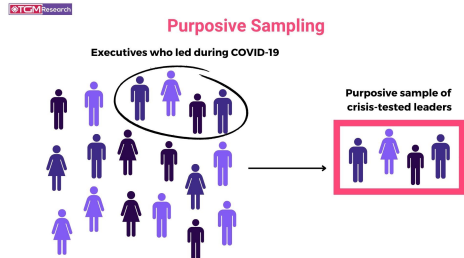
Researcher selects sample based on expertise or judgment.

Example

Selecting subject experts for policy evaluation.

Limitation

Subjective and may introduce bias.



Definition

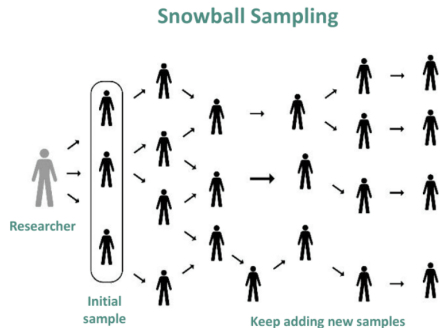
Existing participants recruit future participants.

Used For

- Rare populations
- Sensitive groups

Limitation

Network bias may occur.



Feature	Stratified	Cluster
Group Type	Homogeneous	Heterogeneous
Sampling	From each stratum	Entire cluster selected
Precision	High	Moderate
Cost	Higher	Lower

Guidelines

- Complete list available → **Simple Random**
- Ordered list → **Systematic**
- Population has subgroups → **Stratified**
- Large area → **Cluster**
- Exploratory research → **Convenience**

Exam Tip

Probability sampling is essential for:

- Confidence intervals
- Hypothesis testing

Question

Population size = 2000

Required sample size = 200

Find sampling interval for systematic sampling.

Question

Population size = 2000

Required sample size = 200

Find sampling interval for systematic sampling.

$$k = \frac{N}{n} = \frac{2000}{200} = 10$$

Answer

Select every **10th** unit after a random start.

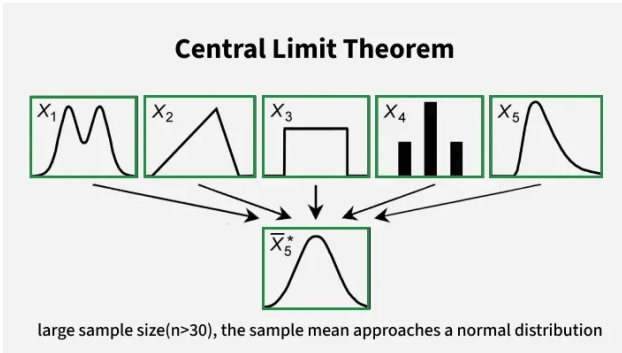
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Central Limit Theorem (CLT)

If $X_1, X_2, \dots, X_n$ are i.i.d. random variables with finite mean μ and finite variance σ^2 , then for large n , the sample mean $\bar{X}$ is approximately normally distributed with mean μ and variance $\frac{\sigma^2}{n}$, regardless of the original population distribution.

Important Remarks

- Valid for large sample size n .
- Requires finite mean and variance.
- Approximation improves as n increases.



Theoretical Idea

- Consider repeated samples from any population.
- Compute the sample mean each time.
- The distribution of these sample means tends toward a **Normal distribution**.
- This happens even if the original population is not normal.

Key Insight

Large samples $\Rightarrow$ sampling distribution of $\bar{X}$ becomes approximately normal.

CLT is an approximation — exact only when population is normal.

Mathematical Statement

Let $X_1, \dots, X_n$ be i.i.d. with

$$E[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty$$

Define:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Then:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad (\text{approx.})$$

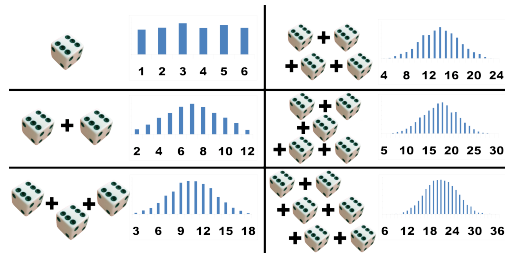
Definition

Random variables $X_1, X_2, \dots, X_n$ are called **i.i.d.** if:

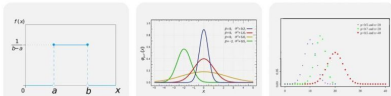
- They are **independent** (no influence on each other),
- They have the **same probability distribution**.

Example

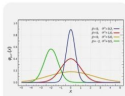
If coin toss results $X_i \sim \text{Bernoulli}(0.5)$ for all i , then the tosses are i.i.d.



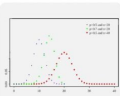
The Most Important Distributions to know in Data Science



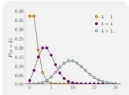
Uniform Distribution



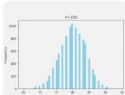
Normal Distribution



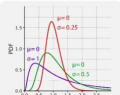
Binomial Distribution



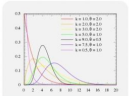
Poisson Distribution



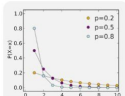
Bernoulli Distribution



Log Normal Distribution



Gamma Distribution



Geometric Distribution



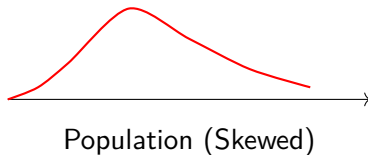
Beta Distribution

Continuous Distributions

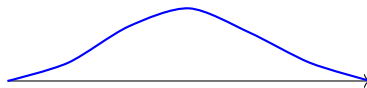
- **Uniform Distribution:** Equal probability over a fixed interval.
- **Normal Distribution:** Symmetric bell-shaped distribution characterized by mean and variance.
- **Log-Normal Distribution:** Logarithm of the variable follows a normal distribution.
- **Gamma Distribution:** Flexible distribution used in waiting-time and reliability modeling.
- **Beta Distribution:** Defined on $(0, 1)$, used for modeling probabilities.

Discrete Distributions

- **Bernoulli Distribution:** Single trial with success/failure outcome.
- **Binomial Distribution:** Number of successes in repeated independent trials.
- **Geometric Distribution:** Number of trials until first success.
- **Poisson Distribution:** Counts number of events in a fixed interval.

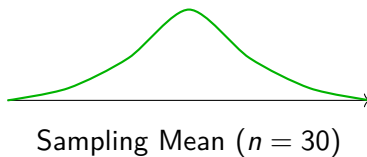


Distribution becomes increasingly symmetric and Gaussian.



Sampling Mean ($n = 5$)

Distribution becomes increasingly symmetric and Gaussian.



Distribution becomes increasingly symmetric and Gaussian.



Distribution becomes increasingly symmetric and Gaussian.

Question:

The waiting time at a service center is highly skewed with mean 30 minutes and SD 10 minutes.

For $n = 100$, find:

$$P(\bar{X} < 32)$$

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For $n = 100$, find:

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Step 1: Given

$$\mu = 30, \quad \sigma = 10, \quad n = 100$$

Step 2: Apply CLT

$$\bar{X} \sim N\left(30, \frac{10^2}{100}\right)$$

$$SE = \frac{10}{10} = 1$$

Question:

The waiting time at a service center is highly skewed with mean 30 minutes and SD 10 minutes.
For $n = 100$, find:

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Step 2: Apply CLT

$$\bar{X} \sim N\left(30, \frac{10^2}{100}\right)$$

$$SE = \frac{10}{10} = 1$$

Step 3: Standardize

$$Z = \frac{32 - 30}{1} = 2$$

Step 4: Probability

$$P(Z < 2) = 0.9772$$

Decision:

$$P(\bar{X} < 32) = 0.9772$$

Question:

The waiting time at a service center is highly skewed with mean 30 minutes and SD 10 minutes.
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Even though population is skewed, CLT gives normal approximation.

Question:

Population mean = 80, variance = 100. For $n = 36$, find:

$$P(78 < \bar{X} < 82)$$

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Population mean = 80, variance = 100. For $n = 36$, find:

$$P(78 < \bar{X} < 82)$$

Step 1:

$$\sigma = 10, \quad n = 36$$

$$SE = \frac{10}{6} = 1.667$$

$$\bar{X} \sim N(80, 2.78)$$

Question:

Population mean = 80, variance = 100. For $n = 36$, find:

$$P(78 < \bar{X} < 82)$$

Step 1:

$$\sigma = 10, \quad n = 36$$

$$SE = \frac{10}{6} = 1.667$$

$$\bar{X} \sim N(80, 2.78)$$

Final Probability = 0.7698

Step 2:

$$Z_1 = \frac{-2}{1.667} = -1.2$$

$$Z_2 = 1.2$$

Step 3:

$$P(-1.2 < Z < 1.2) = 0.7698$$

Question:

A population has mean 120 and standard deviation 30. For $n = 49$, find:

$$P(\bar{X} > 125)$$

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A population has mean 120 and standard deviation 30. For $n = 49$, find:

$$P(\bar{X} > 125)$$

Step 1:

$$\mu = 120, \quad \sigma = 30, \quad n = 49$$

$$SE = \frac{30}{7} = 4.286$$

$$\bar{X} \sim N(120, 18.37)$$

Question:

A population has mean 120 and standard deviation 30. For $n = 49$, find:

$$P(\bar{X} > 125)$$

Step 1:

$$\mu = 120, \quad \sigma = 30, \quad n = 49$$

$$SE = \frac{30}{7} = 4.286$$

$$\bar{X} \sim N(120, 18.37)$$

Step 2:

$$Z = \frac{125 - 120}{4.286}$$

$$Z = 1.17$$

Question:

A population has mean 120 and standard deviation 30. For $n = 49$, find:

$$P(\bar{X} > 125)$$

Step 1:

$$\mu = 120, \quad \sigma = 30, \quad n = 49$$

$$SE = \frac{30}{7} = 4.286$$

$$\bar{X} \sim N(120, 18.37)$$

Final Probability = 0.1210

Step 2:

$$Z = \frac{125 - 120}{4.286}$$

$$Z = 1.17$$

Step 3:

$$P(Z > 1.17) = 0.1210$$

Question:

The average weekly expenditure is 2000 with SD 500. For $n = 100$, find:

$$P(|\bar{X} - 2000| < 100)$$

Question:

The average weekly expenditure is 2000 with SD 500. For $n = 100$, find:

$$P(|\bar{X} - 2000| < 100)$$

Step 1:

$$\mu = 2000, \quad \sigma = 500, \quad n = 100$$

$$SE = \frac{500}{10} = 50$$

Probability = 0.9544

Step 2:

$$Z = \frac{100}{50} = 2$$

Step 3:

$$P(-2 < Z < 2) = 0.9544$$

Question:

The mean height is 170 cm with SD 6 cm. For $n = 64$, find:

$$P(\bar{X} < 168)$$

Question:

The mean height is 170 cm with SD 6 cm. For $n = 64$, find:

$$P(\bar{X} < 168)$$

Step 1:

$$\mu = 170, \quad \sigma = 6, \quad n = 64$$

$$SE = \frac{6}{8} = 0.75$$

Step 2:

$$Z = \frac{168 - 170}{0.75}$$

$$Z = -2.67$$

Step 3:

$$P(Z < -2.67) = 0.0038$$

Very small probability

Question:

A highly skewed distribution has mean 50 and SD 20. For $n = 100$, find:

$$P(48 < \bar{X} < 52)$$

Question:

A highly skewed distribution has mean 50 and SD 20. For $n = 100$, find:

$$P(48 < \bar{X} < 52)$$

Step 1:

$$\mu = 50, \quad \sigma = 20, \quad n = 100$$

$$SE = \frac{20}{10} = 2$$

Step 2:

$$Z_1 = \frac{-2}{2} = -1$$

$$Z_2 = 1$$

Step 3:

$$P(-1 < Z < 1) = 0.6826$$

CLT applies despite skewness

Question:

A coin has $p = 0.6$. It is tossed 200 times.

Find:

$$P(0.55 < \hat{p} < 0.65)$$

Question:

A coin has $p = 0.6$. It is tossed 200 times.

Find:

$$P(0.55 < \hat{p} < 0.65)$$

Step 1:

$$\mu_{\hat{p}} = p = 0.6$$

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

$$= \sqrt{\frac{0.6(0.4)}{200}}$$

$$= 0.0346$$

Step 2:

$$Z_1 = \frac{-0.05}{0.0346} = -1.45$$

$$Z_2 = 1.45$$

Step 3:

$$P(-1.45 < Z < 1.45) = 0.853$$

Probability 0.853

Question:

Daily demand has mean 200 and SD 30. For 50 days, find:

$$P(S_{50} > 10500)$$

Question:

Daily demand has mean 200 and SD 30. For 50 days, find:

$$P(S_{50} > 10500)$$

Step 1:

$$E(S_n) = n\mu = 50(200) = 10000$$

$$\text{Var}(S_n) = n\sigma^2 = 50(900)$$

$$\sigma_{S_n} = \sqrt{45000} = 212.13$$

Very small probability of exceeding 10500

Step 2:

$$Z = \frac{10500 - 10000}{212.13}$$

$$Z = 2.36$$

Step 3:

$$P(Z > 2.36) = 0.0091$$

Question:

Mean = 50, SD = 15. Find n such that:

$$P(|\bar{X} - 50| < 2) > 0.95$$

Question:

Mean = 50, SD = 15. Find n such that:

$$P(|\bar{X} - 50| < 2) > 0.95$$

Step 1:
95

$$\frac{2}{SE} = 1.96$$

$$SE = \frac{2}{1.96} = 1.02$$

Minimum sample size 216

Step 2:

$$\frac{15}{\sqrt{n}} = 1.02$$

$$\sqrt{n} = 14.7$$

$$n = 216$$

Failure Conditions

- **Infinite variance**
- Extremely heavy tails
- Strong dependence between variables

Example: Cauchy Distribution

$E[X]$ does not exist

$$\text{Var}(X) = \infty$$

What Goes Wrong?

- Sample mean does **not converge**.
- No variance shrinkage.
- No normal limiting distribution.

Dependence Issue

If observations are correlated,

$$\text{Var}(\bar{X}) \neq \frac{\sigma^2}{n}$$

Classical CLT formula breaks.

Classical CLT requires finite variance and independence.

- 1 Inferential Statistics
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- 12 F-TEST
- 13 Analysis of Variance (ANOVA)

Core Question

Is the observed difference real or due to random sampling variation?

- We rarely observe the entire population.
- We collect a sample.
- Sample results vary due to randomness.
- We need a formal decision framework.

Goal

Make statistically valid decisions about population parameters.

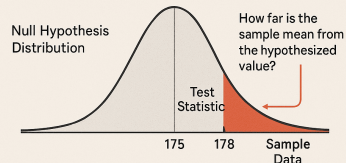
Understanding Test Statistics: A Comprehensive Guide

What is a Hypothesis Test

Null Hypothesis (H_0)
Population mean = 175

Alternative Hypothesis
Population mean $\neq$ 175

The Role of the Test Statistic



$$\text{Test Statistic} = \frac{\text{Sample Statistic} - \text{Population}}{\text{Standard Error}}$$

$$t = \frac{178 - 175}{s / \sqrt{n}} \quad t = 3.29$$

Interpreting the Test Statistic

- Compare to a Critical Value
- Calculate a p-value

Population

- Entire group of interest
- True parameters:
 - Mean (μ)
 - Variance (σ^2)
 - Proportion (p)

Sample

- Subset of population
- Sample statistics:
 - Mean ($\bar{x}$)
 - Variance (s^2)
 - Sample proportion ($\hat{p}$)

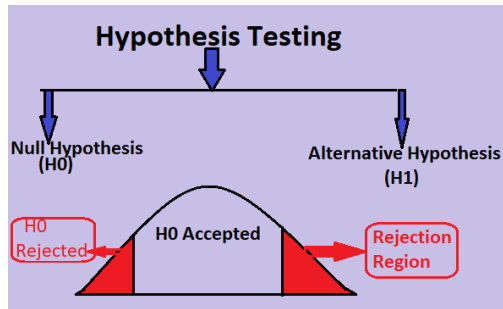
Inference

Use sample statistics to estimate population parameters.

Definition

A hypothesis is a claim about a population parameter.

- **Null Hypothesis (H_0):**
 - Status quo
 - No effect / No difference
- **Alternative Hypothesis (H_1):**
 - Competing claim
 - There is an effect or difference



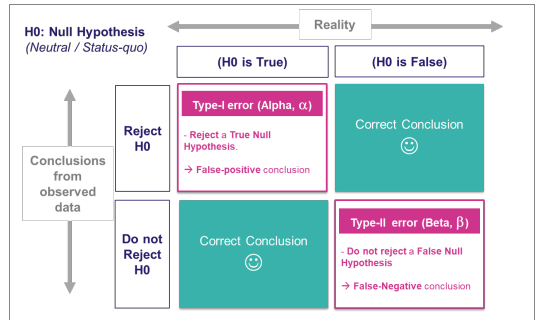
$$H_0 : \mu = \mu_0$$

$$H_1 : \mu \neq \mu_0$$

Two Types of Errors

Reality	Decision	Result
H_0 true	Reject H_0	Type I Error
H_0 false	Do not reject H_0	Type II Error

- Type I Error probability = α
- Type II Error probability = β
- Power of test = $1 - \beta$



Significance Level (α)

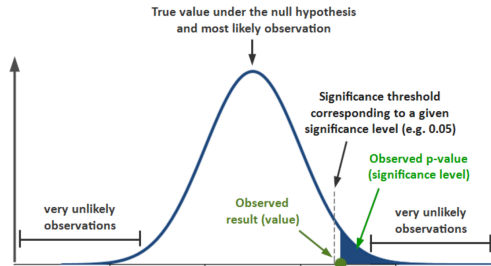
Maximum probability of committing Type I error.

Common choices:

$$\alpha = 0.05, \quad 0.01, \quad 0.10$$

Interpretation

$\alpha = 0.05$ means: We allow 5% risk of rejecting a true null hypothesis.



Right-Tailed Test

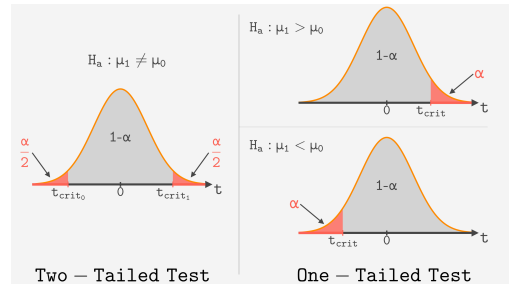
$$H_1 : \mu > \mu_0$$

Left-Tailed Test

$$H_1 : \mu < \mu_0$$

Two-Tailed Test

$$H_1 : \mu \neq \mu_0$$



General Idea

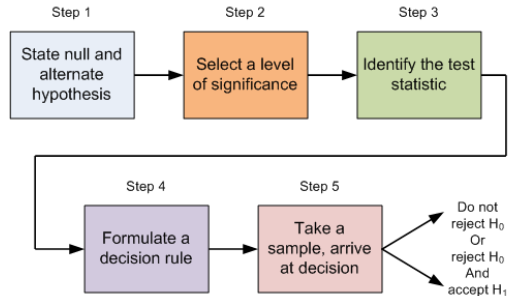
Convert sample information into a standardized value.

$$\text{Test Statistic} = \frac{\text{Observed Value} - \text{Expected Value}}{\text{Standard Error}}$$

- Measures deviation from null hypothesis
- Larger deviation $\Rightarrow$ stronger evidence against H_0

- 1 State H_0 and H_1
- 2 Choose α
- 3 Select appropriate test statistic
- 4 Compute test statistic
- 5 Determine p-value or critical value
- 6 Make statistical decision
- 7 Interpret in context

Five-Step Procedure for Testing a Hypothesis



- A/B testing
- Model comparison
- Feature selection
- ANOVA
- Testing regression coefficients
- Independence testing

Key Insight

Hypothesis testing is a probabilistic decision-making framework based on sampling variability.

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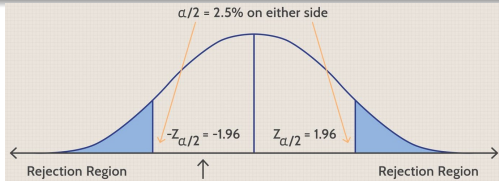
Objective

The Z-test is a **parametric hypothesis test** used to determine whether a sample provides sufficient evidence to reject a claim about a population mean (or proportion), when population variance is known.

Population vs Sample

- Population Mean: μ
- Sample Mean: $\bar{X}$
- Population SD: σ
- Sample Size: n

Inference is required because μ is unknown.



Sampling Distribution of Mean

If:

- Population is normal, OR
- $n \geq 30$ (CLT)

Then:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$E(\bar{X}) = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

$$SE = \frac{\sigma}{\sqrt{n}}$$

Standardization Principle

To measure deviation from the claimed mean:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

This converts the sampling distribution into: $Z \sim N(0, 1)$

Interpretation

Z represents:

Number of Standard Errors the sample mean is away from μ

Hypothesis Framework

Null Hypothesis (H_0):

$$\mu = \mu_0$$

Alternative Hypothesis (H_1):

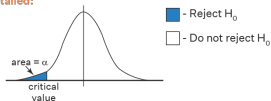
$$\mu \neq \mu_0 \quad (\text{Two-tailed})$$

$$\mu > \mu_0 \quad (\text{Right-tailed})$$

$$\mu < \mu_0 \quad (\text{Left-tailed})$$

Rejection Region for Null Hypothesis

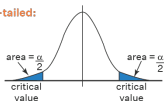
left-tailed:



right-tailed:



two-tailed:



Level of Significance (α)

Probability of Type I Error:

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ true})$$

Common values: $\alpha = 0.10, 0.05, 0.01$

Decision Rule

Compute: $Z_{cal} = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$

Reject H_0 if: $|Z_{cal}| > Z_{critical}$

Otherwise: Fail to reject H_0

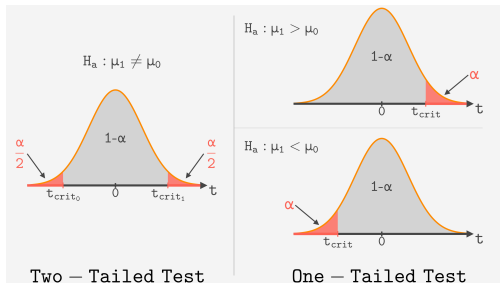
Geometric Interpretation: Large $|Z|$ implies the observed sample mean lies in the low-probability tail region under H_0 , indicating statistical significance.

Critical Values (Two-Tailed)

$$\alpha = 0.10 \Rightarrow Z = \pm 1.645$$

$$\alpha = 0.05 \Rightarrow Z = \pm 1.96$$

$$\alpha = 0.01 \Rightarrow Z = \pm 2.576$$



Critical Values (One-Tailed Test)

Right-Tailed Test: Reject H_0 if $Z > Z_\alpha$

$$\alpha = 0.10 \Rightarrow Z = 1.28$$

$$\alpha = 0.05 \Rightarrow Z = 1.645$$

$$\alpha = 0.01 \Rightarrow Z = 2.33$$

Left-Tailed Test: Reject H_0 if $Z < -Z_\alpha$

$$\alpha = 0.10 \Rightarrow Z = -1.28$$

$$\alpha = 0.05 \Rightarrow Z = -1.645$$

$$\alpha = 0.01 \Rightarrow Z = -2.33$$

Z-Test for Proportion

If:

$$\hat{p} = \frac{x}{n}$$

Then:

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

Used for inference on population proportions.

Role in Data Science

- Foundation of hypothesis testing
- Basis for regression inference
- Model validation logic
- Precursor to ANOVA and advanced inference

Problem

A coaching institute claims the average score is 75. A sample of 40 students has mean 71. Population SD = 5. Test at 5% level.

Problem

A coaching institute claims the average score is 75. A sample of 40 students has mean 71. Population SD = 5. Test at 5% level.

Claimed population mean: $\mu_0 = 75$

Sample size: $n = 40$

Sample mean: $\bar{X} = 71$

Population standard deviation: $\sigma = 5$

Level of significance: $\alpha = 0.05$

Problem

A coaching institute claims the average score is 75. A sample of 40 students has mean 71. Population SD = 5. Test at 5% level.

Claimed population mean: $\mu_0 = 75$

Sample size: $n = 40$

Sample mean: $\bar{X} = 71$

Population standard deviation: $\sigma = 5$

Level of significance: $\alpha = 0.05$

Step 1: Hypotheses

$$H_0 : \mu = 75 \quad ; \quad H_1 : \mu \neq 75$$

Problem

A coaching institute claims the average score is 75. A sample of 40 students has mean 71. Population SD = 5. Test at 5% level.

Claimed population mean: $\mu_0 = 75$

Sample size: $n = 40$

Sample mean: $\bar{X} = 71$

Population standard deviation: $\sigma = 5$

Level of significance: $\alpha = 0.05$

Step 1: Hypotheses

$$H_0 : \mu = 75 \quad ; \quad H_1 : \mu \neq 75$$

Step 2: Test Statistic

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{71 - 75}{5/\sqrt{40}} = \frac{-4}{0.79} = -5.06$$

Problem

A coaching institute claims the average score is 75. A sample of 40 students has mean 71. Population SD = 5. Test at 5% level.

Claimed population mean: $\mu_0 = 75$

Sample size: $n = 40$

Sample mean: $\bar{X} = 71$

Population standard deviation: $\sigma = 5$

Level of significance: $\alpha = 0.05$

Step 1: Hypotheses

$$H_0 : \mu = 75 \quad ; \quad H_1 : \mu \neq 75$$

Step 2: Test Statistic

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{71 - 75}{5/\sqrt{40}} = \frac{-4}{0.79} = -5.06$$

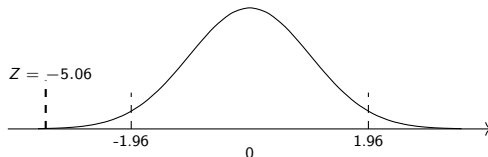
Interpretation: Sample mean lies far in the rejection region.

Step 3: Critical Value

At 5% (two-tailed): ± 1.96

Decision:

$$|Z| = 5.06 > 1.96 \Rightarrow \text{Reject } H_0$$



Conclusion: Mean score significantly differs from 75. We **Reject** the Null Hypothesis.

Problem

A company claims average battery life is 10 hours. Sample: $n = 50$, $\bar{X} = 10.8$, $\sigma = 2$. Test at 5% (Two tail test)

Problem

A company claims average battery life is 10 hours. Sample: $n = 50$, $\bar{X} = 10.8$, $\sigma = 2$. Test at 5% (Two tail test)

Claimed population mean: $\mu_0 = 10$ hours

Sample size: $n = 50$

Sample mean: $\bar{X} = 10.8$

Population standard deviation: $\sigma = 2$

Level of significance: $\alpha = 0.05$

Step 1: State Hypotheses

$$H_0 : \mu = 10 \quad ; \quad H_1 : \mu > 10$$

Step 2: Compute Test Statistic

$$Z = \frac{10.8 - 10}{2/\sqrt{50}} = \frac{0.8}{0.283} = 2.83$$

Problem

A company claims average battery life is 10 hours. Sample: $n = 50$, $\bar{X} = 10.8$, $\sigma = 2$. Test at 5% (Two tail test)

Claimed population mean: $\mu_0 = 10$ hours

Sample size: $n = 50$

Sample mean: $\bar{X} = 10.8$

Population standard deviation: $\sigma = 2$

Level of significance: $\alpha = 0.05$

Step 1: State Hypotheses

$$H_0 : \mu = 10 \quad ; \quad H_1 : \mu > 10$$

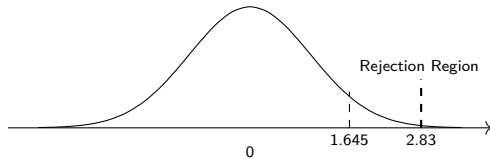
Step 2: Compute Test Statistic

$$Z = \frac{10.8 - 10}{2/\sqrt{50}} = \frac{0.8}{0.283} = 2.83$$

Conclusion: Evidence supports increase in battery life.

Step 3: Critical Value Critical value (5%, right tail): 1.645

$$2.83 > 1.645 \Rightarrow \text{Reject } H_0$$



Interpretation: Test statistic lies in the right-tail rejection region.

Problem

A factory claims 60% items are defect-free. Sample: 200 items, 130 defect-free. Test at 5%.

Problem

A factory claims 60% items are defect-free. Sample: 200 items, 130 defect-free. Test at 5%.

$$\hat{p} = \frac{130}{200} = 0.65$$

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} = \frac{0.65 - 0.60}{\sqrt{\frac{0.6(0.4)}{200}}} = \frac{0.05}{0.0346} = 1.45$$

Critical value (two-tailed, 5%): 1.96

$$1.45 < 1.96 \Rightarrow \text{Fail to reject } H_0$$

Conclusion: No significant change in defect rate.

- ① A hospital claims average patient wait time is 30 minutes. Sample of 36 patients: mean = 28, $\sigma = 6$. Test at 5%.
- ② A machine fills bottles with 500 ml on average. $n = 64$, $\bar{X} = 497$, $\sigma = 8$. Test at 1% (left-tailed).
- ③ A university claims 70% students pass an exam. Out of 400 students, 300 passed. Test at 5%.
- ④ A factory claims mean rod length = 100 cm. $n = 49$, $\bar{X} = 102$, $\sigma = 7$. Test at 10% (two-tailed).

Find: Hypotheses, Z-value, Critical value, Decision.

1.

$$Z = \frac{28 - 30}{6/\sqrt{36}} = \frac{-2}{1} = -2$$

Critical value = ± 1.96 Reject H_0

2.

$$Z = \frac{497 - 500}{8/\sqrt{64}} = \frac{-3}{1} = -3$$

Critical value (1%) = -2.33 Reject H_0

3.

$$\hat{p} = 0.75$$
$$Z = \frac{0.75 - 0.70}{\sqrt{\frac{0.7(0.3)}{400}}} = 2.18$$

Critical value = 1.96 Reject H_0

4.

$$Z = \frac{102 - 100}{7/\sqrt{49}} = \frac{2}{1} = 2$$

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Why do we need t-test?

- In real-world data analysis, **population standard deviation σ is unknown**.
- We estimate variability using **sample standard deviation S** .
- Replacing σ with S introduces additional uncertainty.
- That additional variability leads to a new distribution:

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

Key Insight

When σ is unknown and sample size is small, **Standard Normal $\Rightarrow$ Student's t Distribution**

Z-Test

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

- σ known
- Follows $N(0, 1)$

t-Test

$$t = \frac{\bar{X} - \mu}{S / \sqrt{n}}$$

- σ unknown
- Depends on **degrees of freedom**

Extra variability in denominator $\Rightarrow$ Heavier tails

Foundation

For a normal population:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

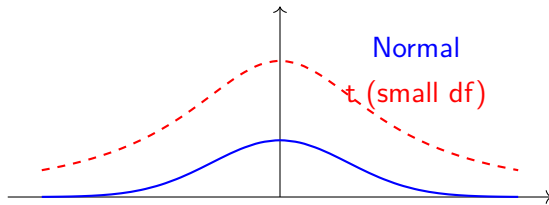
Constructing t

$$t = \frac{Z}{\sqrt{\chi^2/(n-1)}}$$

$$t \sim t_{(n-1)}$$

Interpretation

t-distribution arises from the **ratio of Normal and Chi-square random variables**.

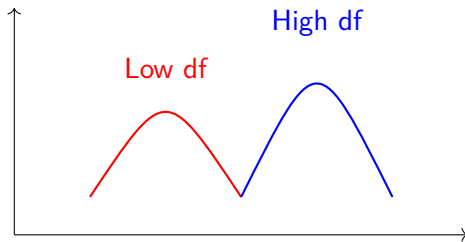


- Symmetric about zero
- Mean = 0
- Heavier tails
- As $df \rightarrow \infty$, $t \rightarrow N(0, 1)$

Definition

$$df = n - 1$$

- Represents number of independent pieces of information.
- One degree lost due to estimation of $\bar{X}$.
- Smaller $df \Rightarrow$ larger variability.



Hypotheses

$$H_0 : \mu = \mu_0$$

$$H_1 : \mu \neq \mu_0 \quad (\text{Two-tailed})$$

Test Statistic

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

- Compare $|t|$ with $t_{\alpha/2, df}$
- Reject H_0 if statistic lies in critical region

1. One-Sample

Tests single population mean.

2. Independent Two-Sample

Compare two independent means.

$$df = n_1 + n_2 - 2$$

3. Paired t-Test

Before-after measurements.

$$t = \frac{\bar{D}}{S_D / \sqrt{n}}$$

Core Idea

All versions measure difference relative to **estimated standard error**.

Feature	Z-Test	t-Test
σ known	Yes	No
Distribution	Normal	t
Depends on df	No	Yes
Tails	Thin	Heavy
Small Sample	Not preferred	Preferred

Practical Rule

If σ unknown $\Rightarrow$ Use t-test

- t-test handles uncertainty in estimating variance.
- Built from Normal and Chi-square distributions.
- Controlled by degrees of freedom.
- Fundamental building block for:
 - ANOVA
 - Regression inference
 - Confidence intervals

t-test = Inference when σ is unknown

Common Significance Levels

df	10% (Two-tail)	5% (Two-tail)	1% (Two-tail)
5	2.571	2.571	4.032
9	1.833	2.262	3.250
15	1.753	2.131	2.947
20	1.725	2.086	2.845
30	1.697	2.042	2.750

Reminder

Right-tail (5%) $\rightarrow t_{0.05, df}$

Two-tail (5%) $\rightarrow t_{0.025, df}$

Problem

A machine is designed to produce rods with mean length 20 cm. A random sample of 16 rods has mean 21 cm and standard deviation 2 cm. Test at 5% significance whether the true mean length differs from 20 cm.

Problem

A machine is designed to produce rods with mean length 20 cm. A random sample of 16 rods has mean 21 cm and standard deviation 2 cm. Test at 5% significance whether the true mean length differs from 20 cm.

$$\mu_0 = 20$$

$$n = 16$$

$$\bar{X} = 21$$

$$S = 2$$

$$\alpha = 0.05$$

Step 1: Hypotheses

$$H_0 : \mu = 20 \quad H_1 : \mu \neq 20$$

Problem

A machine is designed to produce rods with mean length 20 cm. A random sample of 16 rods has mean 21 cm and standard deviation 2 cm. Test at 5% significance whether the true mean length differs from 20 cm.

$$\mu_0 = 20$$

$$n = 16$$

$$\bar{X} = 21$$

$$S = 2$$

$$\alpha = 0.05$$

Step 1: Hypotheses

$$H_0 : \mu = 20 \quad H_1 : \mu \neq 20$$

Step 2: Test Statistic

$$t = \frac{21 - 20}{2/\sqrt{16}} = \frac{1}{0.5} = 2$$

$$df = 15$$

Problem

A machine is designed to produce rods with mean length 20 cm. A random sample of 16 rods has mean 21 cm and standard deviation 2 cm. Test at 5% significance whether the true mean length differs from 20 cm.

$$\mu_0 = 20$$

$$n = 16$$

$$\bar{X} = 21$$

$$S = 2$$

$$\alpha = 0.05$$

Step 1: Hypotheses

$$H_0 : \mu = 20 \quad H_1 : \mu \neq 20$$

Step 2: Test Statistic

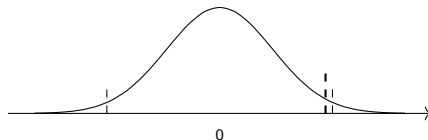
$$t = \frac{21 - 20}{2/\sqrt{16}} = \frac{1}{0.5} = 2$$

$$df = 15$$

Step 3: Critical Value

$$t_{0.025,15} = 2.131$$

$$2 < 2.131 \Rightarrow \text{Fail to Reject } H_0$$



Conclusion: No significant difference. We **fail to reject** Null hypothesis.

Problem

A training institute claims that the average typing speed of its students is 40 words per minute. A random sample of 10 students shows a mean typing speed of 45 wpm with a standard deviation of 5 wpm. At the 5% level of significance, test whether the true mean typing speed is greater than 40 wpm.

Problem

A training institute claims that the average typing speed of its students is 40 words per minute. A random sample of 10 students shows a mean typing speed of 45 wpm with a standard deviation of 5 wpm. At the 5% level of significance, test whether the true mean typing speed is greater than 40 wpm.

$$t_{0.05,9} = 1.833$$

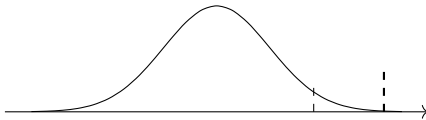
$$H_0 : \mu = 40 \quad H_1 : \mu > 40$$

$$t = \frac{45 - 40}{5/\sqrt{10}} = \frac{5}{1.58} = 3.16$$

df=9

Conclusion: Speed significantly increased.

$$3.16 > 1.833 \Rightarrow \text{Reject } H_0$$



Problem

A nutrition company claims that the mean weight of a certain group is 70 kg. A random sample of 25 individuals has a mean weight of 67 kg and standard deviation 6 kg. At the 5% level of significance, test whether the true mean weight is less than 70 kg.

Problem

A nutrition company claims that the mean weight of a certain group is 70 kg. A random sample of 25 individuals has a mean weight of 67 kg and standard deviation 6 kg. At the 5% level of significance, test whether the true mean weight is less than 70 kg.

$$t_{0.05,24} = 1.711$$

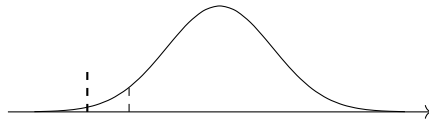
$$H_0 : \mu = 70 \quad H_1 : \mu < 70$$

$$t = \frac{67 - 70}{6/\sqrt{25}} = \frac{-3}{1.2} = -2.5$$

df=24

Conclusion: Mean significantly less.

$$-2.5 < -1.711 \Rightarrow \text{Reject } H_0$$



Problem

A training program is conducted to improve employee performance. The difference in scores (After — Before) for 8 employees has a mean of 4 and a standard deviation of 3. At the 5% level of significance, test whether the training program has significantly improved performance.

$$t_{0.025,7} = 2.365$$

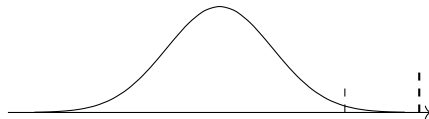
$$H_0 : \mu_D = 0$$

$$t = \frac{4}{3/\sqrt{8}} = \frac{4}{1.06} = 3.77$$

df=7

Conclusion: Training effect significant.

$$3.77 > 2.365 \Rightarrow \text{Reject } H_0$$



1. One-Sample (Two-Tailed)

A population mean is claimed to be 50. Sample: $n = 12$, $\bar{X} = 52$, $S = 4$. Test at $\alpha = 0.05$.

2. One-Sample (Left-Tailed)

A process claims mean output of 100 units. Sample: $n = 18$, $\bar{X} = 98$, $S = 10$. Test at $\alpha = 0.05$ whether mean is less than 100.

3. Paired t-Test

For 9 individuals, the mean difference (After – Before) is $\bar{D} = 2$ with $S_D = 1.5$. Test at $\alpha = 0.01$ whether improvement is significant.

4. Independent Two-Sample Test

Group 1: $n_1 = 10$, $\bar{X}_1 = 60$, $S_1 = 5$

Group 2: $n_2 = 12$, $\bar{X}_2 = 55$, $S_2 = 4$

Test at $\alpha = 0.05$ whether the means differ.

1. Two-Tailed

$$t = 1.73, df = 11$$

$$t_{0.025} = 2.201$$

$$1.73 < 2.201 \Rightarrow \text{Fail to Reject } H_0$$

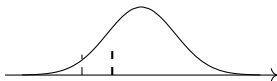


2. Left-Tailed

$$t = -0.85, df = 17$$

$$t_{0.05} = 1.740$$

$$-0.85 > -1.740 \Rightarrow \text{Fail to Reject } H_0$$



3. Paired Test

$$t = 4, df = 8$$

$$t_{0.005} = 3.355$$

$$4 > 3.355 \Rightarrow \text{Reject } H_0$$



4. Two-Sample

$$t = 2.47, df = 20$$

$$t_{0.025} = 2.086$$

$$2.47 > 2.086 \Rightarrow \text{Reject } H_0$$



Visual Rule: If test statistic falls in shaded tail → Reject

- 1 Inferential Statistics
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- 8 Hypothesis Testing
- 9 Z-Test
- 10 Student's t-Test
- 11 Chi-Square (χ^2) Test**
- 12 F-TEST
- 13 Analysis of Variance (ANOVA)

Why χ^2 ?

- Z and t tests focus on **population mean**.
- But what about **population variance**?
- What if we want to test:
 - Whether data follows a theoretical distribution?
 - Whether two categorical variables are independent?

Key Idea

χ^2 measures **how much observed data deviates from expectation**.

Definition

If $Z_1, Z_2, \dots, Z_k \sim N(0, 1)$ independently,

$$\chi^2 = Z_1^2 + Z_2^2 + \dots + Z_k^2$$

follows a Chi-square distribution with k degrees of freedom.

- Sum of squared standard normals
- Always non-negative
- Depends on degrees of freedom

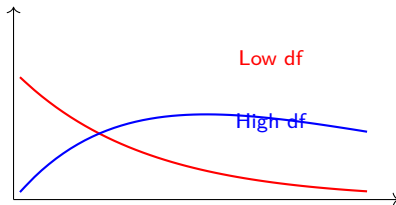
Interpretation

χ^2 quantifies total squared standardized deviation.

Core Properties

- Defined for $x \geq 0$
- Right-skewed
- Mean = df
- Variance = $2df$
- Skewness decreases as df increases

Shape of χ^2 Distribution

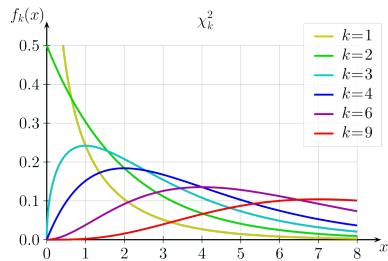


Limiting Behaviour

As $df \rightarrow \infty$:

$$\chi^2 \rightarrow \text{Normal}$$

(Becomes approximately symmetric)



Objective

To test whether the population variance σ^2 is equal to a specified value σ_0^2 .

Hypotheses

$$H_0 : \sigma^2 = \sigma_0^2$$

$$H_1 : \sigma^2 \neq \sigma_0^2 \text{ (or } >, < \text{ depending on problem)}$$

Test Statistic

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$$

- Derived from the fact that $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{(n-1)}^2$.
- Follows Chi-square distribution with $df = n - 1$.
- For variance increase testing $\rightarrow$ Right-tailed test.
- For two-sided variance testing $\rightarrow$ Both tails are considered.
- Larger χ^2 value indicates greater deviation from claimed variance.

Important

Used specifically when making inference about **population variability**, not the mean.

Objective

To test whether the observed categorical frequencies follow a specified theoretical distribution.

Hypotheses

H_0 : Observed frequencies follow the expected distribution

H_1 : Observed frequencies do not follow the distribution

Test Statistic

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- O = Observed frequency
- E = Expected frequency (under H_0)
- Follows $\chi^2_{(k-1)}$
- $df = k - 1$ (k = number of categories)
- Large χ^2 indicates poor fit.

Important

Used to test distributional assumptions such as fairness of dice, coin bias, categorical model validation, etc.

Objective

To test whether two categorical variables are statistically independent.

Hypotheses

H_0 : Variables are independent

H_1 : Variables are associated

Test Statistic

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- O = Observed cell frequency
- $E = \frac{(\text{Row total})(\text{Column total})}{\text{Grand total}}$
- $df = (r - 1)(c - 1)$
- Based on contingency tables
- Larger χ^2 indicates stronger evidence of dependence

Important

Used to detect association between categorical variables such as gender vs performance, smoking vs disease, etc.

Problem

A company claims that the variance of product weight is 16. A sample of 15 items has variance 25. Test at 5% level whether variability has increased.

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A company claims that the variance of product weight is 16. A sample of 15 items has variance 25. Test at 5% level whether variability has increased.

Claimed variance: $\sigma_0^2 = 16$

Sample size: $n = 15$

Sample variance: $S^2 = 25$

$\alpha = 0.05$

Step 1: Hypotheses

$$H_0 : \sigma^2 = 16$$

$$H_1 : \sigma^2 > 16$$

Problem

A company claims that the variance of product weight is 16. A sample of 15 items has variance 25. Test at 5% level whether variability has increased.

Claimed variance: $\sigma_0^2 = 16$

Sample size: $n = 15$

Sample variance: $S^2 = 25$

$\alpha = 0.05$

Step 1: Hypotheses

$$H_0 : \sigma^2 = 16$$

$$H_1 : \sigma^2 > 16$$

Step 2: Test Statistic

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} = \frac{14 \times 25}{16} = 21.88$$

Problem

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$$H_1 : \sigma^2 > 16$$

Step 2: Test Statistic

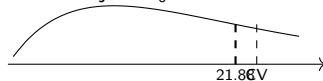
$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} = \frac{14 \times 25}{16} = 21.88$$

Step 3: Critical Value

$$\chi_{0.05,14}^2 = 23.68$$

$$21.88 < 23.68$$

Fail to Reject H_0



Interpretation: Statistic not in rejection region.

Conclusion: No evidence that variance increased.

Solved Example 2 (Chi-Square Variance – Two Tail)

Problem

A manufacturing process is claimed to have a population variance of 36. A random sample of 20 observations yields a sample variance of 20. At the 5% level of significance, test whether the true population variance differs from 36.

Problem

A manufacturing process is claimed to have a population variance of 36. A random sample of 20 observations yields a sample variance of 20. At the 5% level of significance, test whether the true population variance differs from 36.

$$\sigma_0^2 = 36, S^2 = 20, n = 20$$

Step 1: Hypotheses

$$H_0 : \sigma^2 = 36$$

$$H_1 : \sigma^2 \neq 36$$

Step 2: Test Statistic

$$\chi^2 = \frac{19 \times 20}{36} = 10.56$$

$$df = 19$$

Problem

A manufacturing process is claimed to have a population variance of 36. A random sample of 20 observations yields a sample variance of 20. At the 5% level of significance, test whether the true population variance differs from 36.

$$\sigma_0^2 = 36, S^2 = 20, n = 20$$

Step 1: Hypotheses

$$H_0 : \sigma^2 = 36$$

$$H_1 : \sigma^2 \neq 36$$

Step 2: Test Statistic

$$\chi^2 = \frac{19 \times 20}{36} = 10.56$$

$$df = 19$$

Step 3: Critical Values

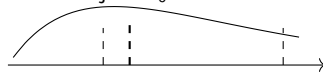
$$\chi_{0.025,19}^2 = 32.85$$

$$\chi_{0.975,19}^2 = 8.91$$

Since:

$$8.91 < 10.56 < 32.85$$

Fail to Reject H_0



Interpretation: Lies between bounds. **Conclusion:** Variance not significantly different.

Problem

A coin is tossed 100 times to examine whether it is fair. The outcomes show 60 heads and 40 tails. At the 5% level of significance, test whether the coin is fair.

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A coin is tossed 100 times to examine whether it is fair. The outcomes show 60 heads and 40 tails. At the 5% level of significance, test whether the coin is fair.

Expected frequencies:

$$E_H = 50, \quad E_T = 50$$

$$\begin{aligned}\chi^2 &= \sum \frac{(O - E)^2}{E} \\ &= \frac{(60 - 50)^2}{50} + \frac{(40 - 50)^2}{50} \\ &= \frac{100}{50} + \frac{100}{50} = 4\end{aligned}$$

$$df = 1$$

Problem

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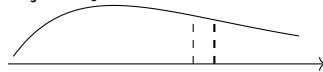
$$df = 1$$

Critical value:

$$\chi_{0.05,1}^2 = 3.84$$

$$4 > 3.84$$

Reject H_0



Interpretation: Coin not fair. **Conclusion:** Significant deviation from fairness.

Problem

The following data show the performance (Pass/Fail) of students classified by gender:

	Pass	Fail
Male	40	20
Female	30	10

At the 5% level of significance, test whether gender and examination performance are independent.

Solved Example (Chi-Square Test of Independence)

Step 1: Hypotheses

H_0 : Independent

H_1 : Associated

Step 2: Totals

Row: 60, 40

Column: 70, 30

Grand Total = 100

Expected Frequencies

$$E_{ij} = \frac{(Row)(Col)}{100}$$

$$E = \begin{bmatrix} 42 & 18 \\ 28 & 12 \end{bmatrix}$$

Step 3: Compute Statistic

$$\begin{aligned} \chi^2 &= \sum \frac{(O - E)^2}{E} \\ &= \frac{(40 - 42)^2}{42} + \frac{(20 - 18)^2}{18} \\ &\quad + \frac{(30 - 28)^2}{28} + \frac{(10 - 12)^2}{12} \\ &= 0.095 + 0.222 + 0.143 + 0.333 \\ \chi^2 &= 0.79 \end{aligned}$$

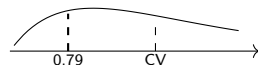
Step 4: Decision

$$df = (2 - 1)(2 - 1) = 1$$

$$\chi^2_{0.05,1} = 3.84$$

$$0.79 < 3.84$$

Fail to Reject H_0



Conclusion: No significant association.

1. A manufacturing process is claimed to have a population variance of 20. A random sample of 18 observations yields a sample variance of 30. At the 5% level of significance, test whether the population variance exceeds 20.

2. A coin is tossed 100 times to examine whether it is fair. The results show 60 heads and 40 tails. At the 5% level of significance, test whether the coin is fair.

3. The following data represent examination results (Pass/Fail) classified by gender:

	Pass	Fail
Male	40	20
Female	30	10

At the 5% level of significance, test whether gender and examination performance are independent.

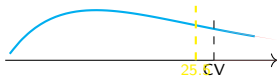
4. A process is claimed to have a population variance of 4. A random sample of 12 observations yields a sample variance of 9. At the 5% level of significance, test whether the true population variance differs from 4.

1. Variance Test (Right-Tailed)

$$\chi^2 = 25.5 \quad df = 17$$

Critical value: 27.59, $25.5 < 27.59$

Decision: Fail to Reject H_0

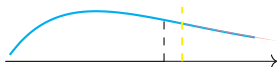


2. Goodness of Fit

$$\chi^2 = 4 \quad df = 1$$

Critical value: 3.84, $4 > 3.84$

Decision: Reject H_0

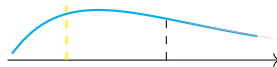


3. Independence Test

$$\chi^2 = 0.79 \quad df = 1$$

Critical value: 3.84, $0.79 < 3.84$

Decision: Fail to Reject H_0

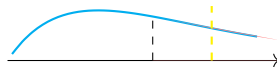


4. Variance Test (Two-Tailed)

$$\chi^2 = 24.75 \quad df = 11$$

Upper CV: 21.92, $24.75 > 21.92$

Decision: Reject H_0



Visual Rule: Right-Shaded Region = Rejection Region

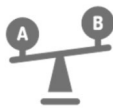
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- 12 F-TEST**
- 13 Analysis of Variance (ANOVA)

Core Question

When do we compare **variability** instead of comparing means?

- Z-test → Compare means (large sample)
- t-test → Compare means (small sample)
- χ^2 → Test single variance
- **F-test → Compare two variances**

What is F-Test?



F-Test

The F-Test is employed to assess the equality of variances between two populations

It is also referred to as variance ratio test as it determines the ratio of variances

In the F-Test, the F-value is compared to the F-critical value

Central Idea

F-test examines whether two populations have equal variability.

Example 1: Coaching Institutes

Two institutes both claim:

- Average marks = 75

But:

- Institute A: Marks tightly clustered
- Institute B: Marks highly scattered

Which is more consistent?

Example 2: Manufacturing

Two machines produce bolts:

- Same average length
- Different variability

Higher variance → More defects.

Example 1: Coaching Institutes

Two institutes both claim:

- Average marks = 75

But:

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Which is more consistent?

Example 2: Manufacturing

Two machines produce bolts:

- Same average length
- Different variability

Higher variance → More defects.

Insight

Even if means are equal, **spread** may differ.

Interpretation

Variance measures the average squared deviation from the mean.

- Large variance → Data widely spread
- Small variance → Data tightly packed

Key Question

If two samples have variances S_1^2 and S_2^2 , are they statistically different?

Formal Definition

The F-test is used to test whether two population variances are equal.

- $S_1^2 \rightarrow$ Larger sample variance
- $S_2^2 \rightarrow$ Smaller sample variance
- Ensures $F \geq 1$

Nature of Test

Right-tailed test only.

F-Test Formula

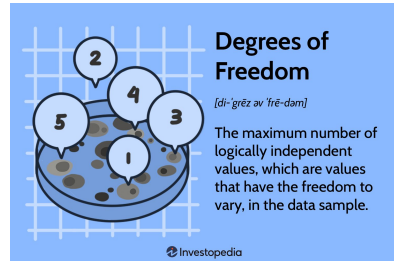


$$\text{F Value} = \frac{\text{Larger Sample Variance}}{\text{Smaller Sample Variance}} = \frac{\sigma_1^2}{\sigma_2^2}$$

Important Property

F-distribution has TWO degrees of freedom.

- $df_1 \rightarrow$ Numerator degrees of freedom
- $df_2 \rightarrow$ Denominator degrees of freedom
- Shape depends on both



- Always positive
- Right skewed
- Not symmetric
- Shape changes with (df_1, df_2)

Important

As degrees of freedom increase, distribution becomes less skewed.

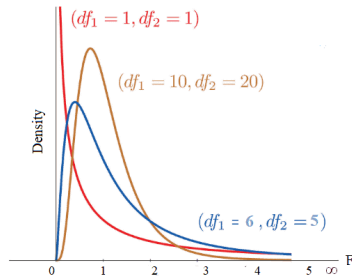


Figure: Illustrative right-skewed shape

Null Hypothesis

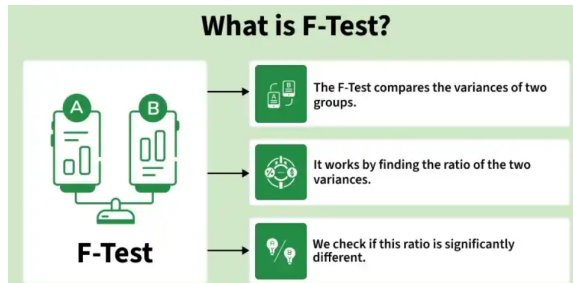
$$H_0 : \sigma_1^2 = \sigma_2^2$$

Alternative Hypothesis

$$H_1 : \sigma_1^2 \neq \sigma_2^2$$

Interpretation

If calculated F is very large, variances are significantly different.



Decision Rule

If $F_{calculated} > F_{critical} \rightarrow \text{Reject } H_0$

- Large ratio $\rightarrow$ Evidence against equality
- Ratio near 1 $\rightarrow$ Variances likely equal

Statistical Meaning

We compare relative variability, not absolute difference.

Derivation Insight

F-distribution arises from ratio of two Chi-square variables.

- Both chi-square variables independent
- Each follows normal population assumption

Why Right-Skewed?

Ratio of positive skewed variables $\rightarrow$ Skewed distribution.

Core ANOVA Idea

Compare multiple means using variance ratios.

- If between variance » within variance
- Means are significantly different

Key Insight

F-test is the mathematical backbone of ANOVA.

Model Significance Test

In regression:

- Tests overall model significance
- Used in linear regression
- Checks whether predictors jointly explain variation

Important for ML

F-test connects classical statistics to machine learning evaluation.

Testing Framework Summary

- $Z \rightarrow$ Compare mean (large sample)
- $t \rightarrow$ Compare mean (small sample)
- $\chi^2 \rightarrow$ Test single variance
- **F $\rightarrow$ Compare two variances**
- ANOVA $\rightarrow$ Compare multiple means

Conceptual Ladder

Mean $\rightarrow$ Variance $\rightarrow$ Ratio of Variances $\rightarrow$ Multi-Group Comparison

Objective

To test whether two population variances are equal.

Step 1: State the Hypotheses

$$H_0 : \sigma_1^2 = \sigma_2^2$$

$$H_1 : \sigma_1^2 \neq \sigma_2^2$$

Step 2: Convert Given Data (If Required)

If standard deviation is given:

$$S^2 = s^2$$

If variance is calculated using divisor n (biased variance):

$$S^2 = \frac{n}{n-1} \times s^2$$

(Always use unbiased variance in F-test.)

Step 3: Calculate the Test Statistic

$$F = \frac{S_1^2}{S_2^2}$$

- Place larger variance in numerator.
- This ensures $F \geq 1$.

Step 4: Degrees of Freedom

$$df_1 = n_1 - 1$$

$$df_2 = n_2 - 1$$

Step 5: Obtain Critical Value

From F-table using:

- Significance level α
- df_1 (numerator degrees of freedom)
- df_2 (denominator degrees of freedom)

Step 6: Decision Rule

Reject H_0 if $F_{calculated} > F_{critical}$

Otherwise, Fail to Reject H_0

(Note: F-test is always right-tailed.)

Final Conclusion (How Students Must Write)

- Since $F_{calculated}$ is (greater/less) than $F_{critical}$,
- We (reject/fail to reject) H_0 .
- Therefore, there is (significant/no significant) difference between the population variances.

Common Mistakes

- Forgetting to square standard deviation.
- Not correcting biased variance (using divisor n instead of $n - 1$).
- Placing smaller variance in numerator.
- Mixing numerator and denominator degrees of freedom.
- Reading wrong value from F-table.
- Writing incorrect conclusion statement.

Question

Two machines produce metal rods. Sample results:

Machine A: $n_1 = 10$, variance $S_1^2 = 25$ Machine B: $n_2 = 12$, variance $S_2^2 = 10$

Test at 5% level whether the population variances differ.

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Two machines produce metal rods. Sample results:

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$$H_0 : \sigma_1^2 = \sigma_2^2 \quad H_1 : \sigma_1^2 \neq \sigma_2^2$$

Step 2: Test Statistic

$$F = \frac{25}{10} = 2.5$$

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Step 2: Test Statistic

$$F = \frac{25}{10} = 2.5$$

Step 3: Degrees of Freedom

$$df_1 = 9, \quad df_2 = 11$$

From F-table at 5%: $F_{critical} = 3.29$

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Step 4: Decision

Since $2.5 < 3.29$, fail to reject H_0 .

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From F-table at 5%: $F_{critical} = 3.29$

Step 4: Decision

Since $2.5 < 3.29$, fail to reject H_0 .

Conclusion: No significant difference in variability.

Question

Two classes took the same test.

Class A: $n_1 = 15$, $S_1^2 = 36$ Class B: $n_2 = 18$, $S_2^2 = 16$

Test equality of variances at 1% significance level.

Question

Two classes took the same test.

Class A: $n_1 = 15$, $S_1^2 = 36$ Class B: $n_2 = 18$, $S_2^2 = 16$

Test equality of variances at 1% significance level.

Solution

Hypotheses

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Class A: $n_1 = 15$, $S_1^2 = 36$ Class B: $n_2 = 18$, $S_2^2 = 16$

Test equality of variances at 1% significance level.

Solution

Hypotheses

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad H_1 : \sigma_1^2 \neq \sigma_2^2$$

Test Statistic

$$F = \frac{36}{16} = 2.25$$

Question

Two classes took the same test.

Class A: $n_1 = 15$, $S_1^2 = 36$ Class B: $n_2 = 18$, $S_2^2 = 16$

Test equality of variances at 1% significance level.

Solution

Hypotheses

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad H_1 : \sigma_1^2 \neq \sigma_2^2$$

Test Statistic

$$F = \frac{36}{16} = 2.25$$

Degrees of Freedom

$$df_1 = 14, \quad df_2 = 17$$

From F-table at 1%: $F_{critical} = 4.17$

Question

Two classes took the same test.

Class A: $n_1 = 15$, $S_1^2 = 36$ Class B: $n_2 = 18$, $S_2^2 = 16$

Test equality of variances at 1% significance level.

Solution

Hypotheses

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad H_1 : \sigma_1^2 \neq \sigma_2^2$$

Test Statistic

$$F = \frac{36}{16} = 2.25$$

Degrees of Freedom

$$df_1 = 14, \quad df_2 = 17$$

From F-table at 1%: $F_{critical} = 4.17$

Decision

Since $2.25 < 4.17$, fail to reject H_0 .

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Both classes show similar performance consistency.

(Right-tailed rejection region)

Question

Two factories produce batteries.

Factory 1: $n_1 = 8$, $S_1^2 = 49$ Factory 2: $n_2 = 10$, $S_2^2 = 9$

Test at 5% level whether production variability differs.

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Solution

Step 1: Hypotheses

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad H_1 : \sigma_1^2 \neq \sigma_2^2$$

Step 2: Test Statistic

$$F = \frac{49}{9} = 5.44$$

Step 3: Degrees of Freedom

$$df_1 = 7, \quad df_2 = 9$$

From F-table at 5%: $F_{critical} = 3.68$

Question

Two factories produce batteries.

Factory 1: $n_1 = 8$, $S_1^2 = 49$ Factory 2: $n_2 = 10$, $S_2^2 = 9$

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Step 1: Hypotheses

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad H_1 : \sigma_1^2 \neq \sigma_2^2$$

Step 2: Test Statistic

$$F = \frac{49}{9} = 5.44$$

Step 3: Degrees of Freedom

$$df_1 = 7, \quad df_2 = 9$$

From F-table at 5%: $F_{critical} = 3.68$

Step 4: Decision

Since $5.44 > 3.68$, reject H_0 .

Conclusion

Significant difference in production variability.

(Theoretical basis of F distribution)

Question

Two drug formulations are tested for shelf-life variability.

Formulation A: $n_1 = 20$, $S_1^2 = 64$ Formulation B: $n_2 = 22$, $S_2^2 = 25$

Test at 5% level whether variability differs.

Question

Two drug formulations are tested for shelf-life variability.

Formulation A: $n_1 = 20$, $S_1^2 = 64$ Formulation B: $n_2 = 22$, $S_2^2 = 25$

Test at 5% level whether variability differs.

Solution

Hypotheses

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad H_1 : \sigma_1^2 \neq \sigma_2^2$$

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Solution

Hypotheses

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad H_1 : \sigma_1^2 \neq \sigma_2^2$$

Test Statistic

$$F = \frac{64}{25} = 2.56$$

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Two drug formulations are tested for shelf-life variability.

Formulation A: $n_1 = 20$, $S_1^2 = 64$ Formulation B: $n_2 = 22$, $S_2^2 = 25$

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Hypotheses

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Degrees of Freedom

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$$F = \frac{64}{25} = 2.56$$

Degrees of Freedom

$$df_1 = 19, \quad df_2 = 21$$

From F-table at 5%: $F_{critical} = 2.39$

Decision

Since $2.56 > 2.39$, reject H_0 .

Question

Two drug formulations are tested for shelf-life variability.

Formulation A: $n_1 = 20$, $S_1^2 = 64$ Formulation B: $n_2 = 22$, $S_2^2 = 25$

Test at 5% level whether variability differs.

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Degrees of Freedom

$$df_1 = 19, \quad df_2 = 21$$

From F-table at 5%: $F_{critical} = 2.39$

Decision

Since $2.56 > 2.39$, reject H_0 .

Conclusion

Significant difference in shelf-life stability.

(Right-skewed F-distribution)

Question

Two samples are drawn from normal populations.

Sample A: $n_1 = 10$, standard deviation computed using divisor n is $s_1 = 4$.

Sample B: $n_2 = 12$, standard deviation computed using divisor n is $s_2 = 3$.

Test at 5% level whether the population variances differ.

Solution**Step 1: Convert Biased Variance to Unbiased Variance**

Relationship:

$$S^2 = \frac{n}{n-1} \times s^2$$

For Sample A

$$S_1^2 = \frac{10}{9} \times 4^2 = \frac{10}{9} \times 16 = 17.78$$

For Sample B

$$S_2^2 = \frac{12}{11} \times 3^2 = \frac{12}{11} \times 9 = 9.82$$

Solution

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Step 3: Test Statistic

$$F = \frac{17.78}{9.82} = 1.81$$

(Larger variance in numerator)

Solution

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Step 3: Test Statistic

$$F = \frac{17.78}{9.82} = 1.81$$

(Larger variance in numerator)

Step 4: Degrees of Freedom

$$df_1 = 9, \quad df_2 = 11$$

From F-table at 5% level:

$$F_{critical} = 3.29$$

Step 5: Decision

Since $1.81 < 3.29$, fail to reject H_0 .

Final Conclusion

No significant difference between population variances.

Right-tailed F-distribution

Q1: Manufacturing Variability

Machine A: $n_1 = 12$, $s_1 = 5$

Machine B: $n_2 = 15$, $s_2 = 3$

Test at 5% level whether population variances differ.

Q2: Classroom Stability

Class A: $n_1 = 10$, variance (biased) = 20

Class B: $n_2 = 14$, variance (biased) = 12

Test equality of variances at 5%.

Q3: Pharmaceutical Study

Sample 1: $n_1 = 8$, $s_1 = 7$

Sample 2: $n_2 = 9$, $s_2 = 4$

Test at 1% significance level.

Q4: Production Lines

Line A: $n_1 = 20$, $s_1 = 6$

Line B: $n_2 = 18$, $s_2 = 5$

Test at 5% whether variances differ.

Solution Q1

$$S_1^2 = 25, S_2^2 = 9$$

$$F = \frac{25}{9} = 2.78$$

$$df_1 = 11, df_2 = 14$$

$$F_{0.05} = 2.69$$

Since $2.78 > 2.69 \rightarrow$ Reject H_0 .

Significant difference.

Solution Q3

$$S_1^2 = 49, S_2^2 = 16$$

$$F = \frac{49}{16} = 3.06$$

$$df_1 = 7, df_2 = 8$$

$$F_{0.01} = 6.03$$

$3.06 < 6.03 \rightarrow$ Fail to Reject H_0 .

Solution Q2

Unbiased:

$$S_1^2 = \frac{10}{9} \times 20 = 22.22$$

$$S_2^2 = \frac{14}{13} \times 12 = 12.92$$

$$F = 1.72$$

$$df_1 = 9, df_2 = 13$$

$$F_{0.05} = 3.18$$

$1.72 < 3.18 \rightarrow$ Fail to Reject H_0 .

Solution Q4

$$S_1^2 = 36, S_2^2 = 25$$

$$F = 1.44$$

$$df_1 = 19, df_2 = 17$$

$$F_{0.05} = 2.23$$

$1.44 < 2.23 \rightarrow$ No significant difference.

- 1 Inferential Statistics
- 2 Population and Sample
- 3 Quiz: Inferential Statistics
- 4 Sampling Techniques in Inferential Statistics
- 5 Probability Sampling
- 6 Non-Probability Sampling
- 7 Central Limit Theorem
- 8 Hypothesis Testing
- 9 Z-Test
- 10 Student's t-Test
- 11 Chi-Square (χ^2) Test
- 12 F-TEST
- 13 Analysis of Variance (ANOVA)**

Core Question

How do we compare **more than two population means** simultaneously?

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- Two means $\rightarrow$ t-test
- Three or more means $\rightarrow$ ANOVA

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Why Not Multiple t-tests?

Multiple t-tests increase Type I error (inflated false positives).

Core Question

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- Three or more means $\rightarrow$ ANOVA

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Fundamental Idea

Compare variability **between groups** to variability **within groups**.

Real-Life Example

Three teaching methods:

- Traditional
- Smart Classroom
- Hybrid

Do average scores differ?

Real-Life Example

Three teaching methods:

- Traditional
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- Hybrid

Do average scores differ?

Visual Idea

Between-group spread $\uparrow$ Within-group spread $\downarrow$
 $\Rightarrow$ Means likely different

Key Insight

If group means are far apart relative to internal variability $\rightarrow$ Significant difference.

Model Structure

One independent factor with multiple levels.

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Hypotheses

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$$

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Core Statistic

$$F = \frac{\text{Between Group Variance}}{\text{Within Group Variance}}$$

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One independent factor with multiple levels.

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Interpretation

Total Sum of Squares (SST)

$$SST = \sum (x_{ij} - \bar{x})^2$$

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Between Group Sum of Squares (SSB)

$$SSB = \sum n_i (\bar{x}_i - \bar{x})^2$$

Total Sum of Squares (SST)

$$SST = \sum (x_{ij} - \bar{x})^2$$

Between Group Sum of Squares (SSB)

$$SSB = \sum n_i (\bar{x}_i - \bar{x})^2$$

Within Group Sum of Squares (SSW)

$$SSW = \sum (x_{ij} - \bar{x}_i)^2$$

Total Sum of Squares (SST)

$$SST = \sum (x_{ij} - \bar{x})^2$$

Between Group Sum of Squares (SSB)

$$SSB = \sum n_i (\bar{x}_i - \bar{x})^2$$

Within Group Sum of Squares (SSW)

$$SSW = \sum (x_{ij} - \bar{x}_i)^2$$

$$SST = SSB + SSW$$

Source	SS	df	MS
Between	SSB	k-1	$MSB = SSB/(k-1)$
Within	SSW	N-k	$MSW = SSW/(N-k)$

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$$F = \frac{MSB}{MSW}$$

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Between	SSB	k-1	$MSB = SSB/(k-1)$
Within	SSW	N-k	$MSW = SSW/(N-k)$

$$F = \frac{MSB}{MSW}$$

Important

ANOVA compares mean squares (variance estimates).

- Independence of observations
- Normality within each group
- Homogeneity of variances

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Connection to F-Test

Homogeneity of variance is tested using F-test.

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Connection to F-Test

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If Assumptions Fail

Use non-parametric alternatives (Kruskal-Wallis).

What Changes?

Two independent factors affect one dependent variable.

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Two independent factors affect one dependent variable.

Example

- Factor A: Teaching Method
- Factor B: Gender

Do:

- Teaching method affect scores?
- Gender affect scores?
- Interaction effect exist?

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

Where:

- α_i = Effect of Factor A
- β_j = Effect of Factor B
- $(\alpha\beta)_{ij}$ = Interaction effect
- ϵ_{ijk} = Random error

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Where:

- α_i = Effect of Factor A
- β_j = Effect of Factor B
- $(\alpha\beta)_{ij}$ = Interaction effect
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Key Advantage

We can detect interaction between factors.

Total Variation is split into:

- Variation due to Factor A
- Variation due to Factor B
- Interaction Variation
- Random Error

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- Variation due to Factor A
- Variation due to Factor B
- Interaction Variation
- Random Error

Each source has its own:

- Sum of Squares
- Degrees of Freedom
- Mean Square
- F-value

Source	SS	df	F
Factor A	SSA	$a-1$	MSA/MSE
Factor B	SSB	$b-1$	MSB/MSE
Interaction	SSAB	$(a-1)(b-1)$	$MSAB/MSE$
Error	SSE	$ab(n-1)$	-

- Comparing multiple drug treatments
- Comparing multiple machine settings
- Comparing teaching strategies
- A/B/C testing in industry
- Marketing campaign performance
- Hyperparameter tuning in ML

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Data Science Connection

ANOVA helps test whether model performance differs across configurations.

- F-test → Compare two variances
- One-Way ANOVA → Compare multiple means (one factor)
- Two-Way ANOVA → Compare multiple means (two factors + interaction)

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Core Formula

$$F = \frac{\text{Variance Between}}{\text{Variance Within}}$$

- F-test → Compare two variances
- One-Way ANOVA → Compare multiple means (one factor)
- Two-Way ANOVA → Compare multiple means (two factors + interaction)

Core Formula

$$F = \frac{\text{Variance Between}}{\text{Variance Within}}$$

Final Insight

ANOVA transforms mean comparison into variance comparison.

Problem

Three teaching methods are tested on student scores.

Method A: 8, 9, 6 Method B: 5, 7, 4 Method C: 10, 9, 12

At 5% level of significance, test whether the mean scores differ.

Step 1: Hypotheses

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

H_1 : At least one mean differs

Step 2: Compute Means

$$\bar{x}_A = 7.67$$

$$\bar{x}_B = 5.33$$

$$\bar{x}_C = 10.33$$

Grand Mean:

$$\bar{x} = 7.78$$

Step 3: Sum of Squares

$$SSB = 38.88$$

$$SSW = 12.66$$

Step 4: ANOVA Table

Source	SS	df	MS
Between	38.88	2	19.44
Within	12.66	6	2.11

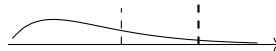
$$F = \frac{19.44}{2.11} = 9.21$$

$$F_{0.05}(2, 6) = 5.14$$

Since $9.21 > 5.14$,

Reject H_0

Conclusion: Means differ significantly.



Problem

A study examines effect of Teaching Method (A, B) and Gender (Male, Female) on test scores.

	Male	Female
Method A	80,85	78,82
Method B	70,75	68,72

Test at 5% level whether:

- Method effect exists
- Gender effect exists
- Interaction effect exists

Solved Example 2 (Two-Way ANOVA with Interaction)

Step 1: Hypotheses

H_0 : No main or interaction effects

Step 2: Compute Means

Grand Mean = 76.25

Method Means:

A = 81.25

B = 71.25

Gender Means:

Male = 77.5

Female = 75

Step 3: Sum of Squares

SSA = 200

SSB = 12.5

SSAB = 2.5

SSE = 25

Step 4: ANOVA Table

Source	SS	df	F
Method	200	1	32
Gender	12.5	1	2
Interaction	2.5	1	0.4
Error	25	4	-

$$F_{0.05}(1, 4) = 7.71$$

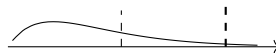
Method: $32 > 7.71 \rightarrow$ Significant

Gender: $2 < 7.71 \rightarrow$ Not Significant

Interaction: $0.4 < 7.71 \rightarrow$ Not Significant

Conclusion:

Only teaching method significantly affects scores.



Concept

An interaction occurs when the effect of one factor depends on the level of another factor.

Example Scenario

Factor A: Teaching Method (A, B)

Factor B: Gender (Male, Female)

Mean Scores:

	Male	Female
Method A	85	78
Method B	75	72

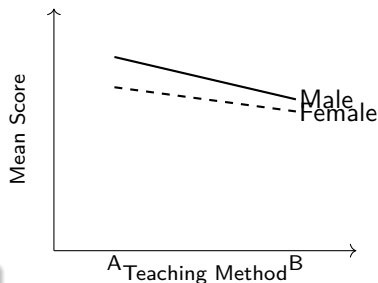
Interpretation Rules

- Parallel lines → No interaction
- Non-parallel lines → Interaction exists
- Crossing lines → Strong interaction

Key Statistical Meaning

If interaction is significant, main effects must be interpreted carefully because factor effects are not independent.

Interaction Plot



Visual Insight:

Lines are not perfectly parallel → mild interaction effect.

Test	Purpose	Data Type	Distribution	df
Z-test	Compare mean	Large sample	Normal (Z)	-
t-test	Compare mean	Small sample	t-distribution	n-1
χ^2 (Variance)	Test single variance	Quantitative	Chi-square	n-1
χ^2 (Independence)	Association test	Categorical	Chi-square	(r-1)(c-1)
F-test	Compare two variances	Quantitative	F-distribution	(n1-1, n2-1)
One-Way ANOVA	Compare >2 means	Quantitative	F-distribution	(k-1, N-k)
Two-Way ANOVA	Multi-factor comparison	Quantitative	F-distribution	Multiple df

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Two-Way ANOVA	Multi-factor comparison	Quantitative	F-distribution	Multiple df

Observation

All major inferential tests are built on comparing estimated variances.

Step 1: What is being tested?

- One mean (large sample) → Z-test
- One mean (small sample) → t-test
- Two means → Independent t-test
- More than two means → ANOVA
- One variance → Chi-square (variance test)
- Two variances → F-test
- Categorical association → Chi-square (independence)

Golden Rule

Mean comparison ultimately becomes variance comparison.

All Tests Follow Same Structure

$$\text{Test Statistic} = \frac{\text{Observed Variation}}{\text{Expected Variation}}$$

- Z-test → Standardized mean difference
- t-test → Standardized mean difference (unknown variance)
- χ^2 → Observed vs Expected counts
- F-test → Ratio of two variances
- ANOVA → Ratio of between and within variances

Core Philosophy

Statistics measures whether observed differences are due to chance.

1. Foundations

- Population vs Sample
- Sampling distributions
- Central Limit Theorem

2. Estimation

- Point estimation
- Confidence intervals

3. Hypothesis Testing Framework

- Null and Alternative hypotheses
- Test statistic
- Critical region / p-value
- Type I and Type II errors

4. Major Tests Covered

- Z-test
- t-test
- Chi-square tests
- F-test
- One-Way ANOVA
- Two-Way ANOVA

Final Takeaway

Inferential statistics helps us move from data to decisions using probability.

- A/B testing in product optimization
- Model comparison in machine learning
- Experimental design
- Feature importance testing
- Scientific research validation

Connection to Machine Learning

- Regression uses F-test
- Model selection uses hypothesis testing
- Variance analysis explains bias-variance tradeoff

Final Insight

Inferential statistics provides the mathematical foundation for evidence-based decision making.

Thank You!

